

LAB EXPERIMENT RECORD

Set 2: Customer Feedback Analysis

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyse customer satisfaction data using a histogram, pie chart and stacked bar chart, and to identify prevalent customer sentiments using a word cloud in R.

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

Procedure

1. Create vectors for customer age and satisfaction score.
2. Use `hist()` to show the distribution of customer ages.
3. Use `table()` to count each satisfaction score and `pie()` to display the distribution with labels.
4. Group ages with `cut()`, build a contingency table with `table()`, and use `barplot()` with `legend()` for the stacked bar chart.
5. Install and load the `wordcloud` package, split the feedback text into words, count them with `table()`, and draw the word cloud using `wordcloud()`.

Key R Functions Used

Function	Purpose
<code>hist()</code>	Creates a histogram of a numeric variable.
<code>table()</code>	Counts frequencies / builds contingency tables.
<code>pie()</code>	Creates a pie chart.
<code>cut()</code>	Converts ages into age groups.
<code>barplot()</code>	Creates the stacked bar chart.
<code>legend()</code>	Adds a legend to a plot.
<code>strsplit()</code> / <code>wordcloud()</code>	Split feedback into words and draw the word cloud.

PROGRAM 1: Histogram of Customer Ages

Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
# Dataset
age <- c(25, 30, 35, 28, 40)

# Histogram
hist(age,
      main = "Distribution of Customer Ages",
      xlab = "Age",
      ylab = "Frequency",
      col = "lightblue")
```



Output 1: Distribution of Customer Ages (Histogram)

Observation: Three of the five customers (ages 25, 28 and 30) fall in the 25–30 range, while one customer each falls in the 30–35 and 35–40 ranges. The customer base is concentrated in the younger age group.

PROGRAM 2: Pie Chart of Satisfaction Scores

Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

```
# Dataset
satisfaction <- c(4, 5, 3, 4, 5)

# Frequency of each score
score_count <- table(satisfaction)

# Pie Chart
pie(score_count,
    labels = names(score_count),
    main = "Customer Satisfaction Distribution",
    col = c("red", "green", "blue"))
```



Output 2: Customer Satisfaction Distribution (Pie Chart)

Observation: Scores 4 and 5 each account for 40% of responses and score 3 accounts for 20%. So 80% of customers gave a score of 4 or above, indicating high overall satisfaction.

PROGRAM 3: Stacked Bar Chart of Satisfaction by Age Group

Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

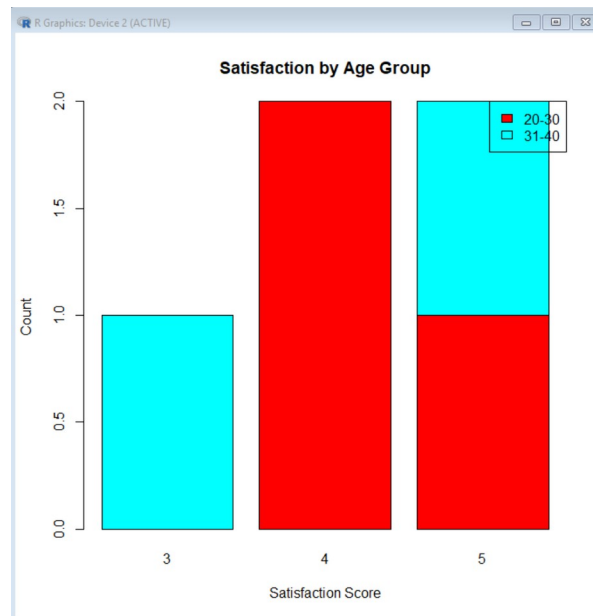
```
# Dataset
age      <- c(25, 30, 35, 28, 40)
satisfaction <- c(4, 5, 3, 4, 5)

# Create age groups
age_group <- cut(age,
                 breaks = c(20, 30, 40),
                 labels = c("20-30", "31-40"))

# Contingency table
data <- table(age_group, satisfaction)

# Stacked Bar Chart
barplot(data,
        col = rainbow(2),
        main = "Satisfaction by Age Group",
        xlab = "Satisfaction Score",
        ylab = "Count")

legend("topright",
      legend = c("20-30", "31-40"),
      fill = rainbow(2))
```



Output 3: Satisfaction by Age Group (Stacked Bar Chart)

Observation: All customers in the 20–30 group gave a score of 4 or 5 (two 4s and one 5). The 31–40 group gave one 3 and one 5, so the only low score comes from the older group. Score 5 is shared equally by both groups.

PROGRAM 4: Word Cloud of Customer Feedback

Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

```
library(wordcloud)

# Open-ended customer feedback
feedback <- c("Excellent product", "Good service", "Fast delivery",
              "Excellent quality", "Good support", "Excellent")

# Split into words and count frequency
words <- unlist(strsplit(tolower(feedback), " "))
word_freq <- table(words)

# Word Cloud
wordcloud(words = names(word_freq),
          freq = as.numeric(word_freq),
          min.freq = 1,
          colors = c("green", "purple"),
          random.order = FALSE)
```



Output 4: Customer Feedback Word Cloud

Observation: "Excellent" (3 mentions) and "good" (2 mentions) are the largest words, showing that customer sentiment is strongly positive. The remaining words — product, quality, service, delivery, fast and support — show which aspects customers are commenting on.

Note: The dataset does not include feedback text, so sample open-ended feedback comments were used to build the word cloud.

Result

The customer satisfaction data was successfully visualized in R using a histogram, pie chart, stacked bar chart and word cloud, and the prevalent customer sentiment was identified as positive.